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Contents

S. No	Title	Page No.
1	Synthesis of silver nanoparticles of some edible basidiomycetes mushroom fungi using response surface methodology and its potential biological application **R Madhanraj, M Eyini and P Balaji**	01
2	Impact of CO_2 on growth, pigments yield and biochemical composition of marine microalga *Dunaliella salina* **A Shenbaga Devi, P Santhanam, S Jeyanthi, B Balaji Prasath and S Dinesh Kumar**	13
3	Fumaronitrile mediated cytochrome P450 (CYP) isoforms biotransformation enzymes responses in *Oreochromis mossambicus* **K Chinnadurai, M Eyini and P Balaji**	23
4	HPLC and biochemical techniques for secondary metabolites in *Garcinia indica* Choisy (Kokum) from transitional zones of Karnataka **Lingappa Sivakumar and Thirugnanasambandam Somasundaram**	35
5	Primary productivity of river chaliyar of Calicut district, Kerala, India **B Dhanalakshmi and P Priyatharsini**	48
6	Anti-bacterial activity, anti-inflammatory and anti- arthritic studies on mangroves by using in vitro model systems **M Babu Selvam and S Abideen**	54
7	Parasitic isopods of the family Cymothoidae from Indian fishes **S Ravichandran and G Ramesh Kumar**	65
8	Isolation and identification of pathogenic bacteria and its antibacterial susceptibility analysis in edible fish *Catla catla* **Mayavan Karthika, Shameem Shabana, Shamoon Muhasin and Venkatachalam Ramasubramanian**	72

9	Biogenic synthesis of silver nanoparticles from *Cardiospermum halicacabum* decorated with Graphene oxide for enhancing antibacterial ability **Gurusamy Sivaprakash, Gujuluva Hari Dinesh, Kulanthaisamy Mohan Rasu, Manoharan Dhivya and Alagarsamy Arun**	80
10	Studies on biosynthesis of xanthan gum using *Xanthomonas* sp., isolated from infected cotton leaves **V Ananthi and A Arun**	88
11	Characterization and determination of antibacterial activity of bacteriocin producing Lactic acid bacteria isolated from curd sample **V Ananthi and A Arun**	95
12	Antibacterial and immunostimulant influence of herbal extracts in grouper *Epinephelus tauvina* experimental culture against *Vibrio harveyi* Infection **T Citarasu, M Michael Babu and SMJ Punitha**	103
13	Assessment of bacteriological quality and presence of antibiotic resistant bacteria in vended sachet-packaged drinking water: potential threat of transmission of enteric pathogens and implications for public health **K Ramamoorthy and Clara G Sargunar**	117
14	Synthesis of chitin form shrimp dispel and its antibacterial activity **P Raja Rajeswari, R Shyamala Gowri, P Meenambigai and K Rajeswari**	132
15	Assessment of antibacterial activity of different solvent extracts of medicinal plant: *Aegle marmelos* **R Shyamala Gowri, R Vijayaraghavan, P Meenambigai and P Raja Rajeswari**	138
16	Effect of aqueous methanolic extract of *Tridax procumbens* on nonspecific immune response of fresh water fish **S Chinniah, T Sangeetha and Subeena Begum**	145
17	A study on biologically synthesize silver nanoparticles using red seaweed *Gracilaria gracilis* **V Veeramanikandan, PT Usha and P Balaji**	154

Synthesis of chitin form shrimp dispel and its antibacterial activity

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Abstract Chitin is one of the most important natural polysaccharide, after cellulose. It is cellulose derivate, it will not occur in organisms producing cellulose. Chitin is annually produced as much as cellulose. The word 'chitin' is referred from the greek word 'chiton'. It is used to determine optimal condition of chitosan production from shrimp by deproteinization, demineralization, and depigmentation, characteristic properties of chitosan. The chitosan is a versatile material with proved as antibacterial activity. The three mechanisms have proposed by antimicrobial activity i) ionic surface interaction ensuing in wall cell leakage ii) inhibition mRNA and protein synthesis by way of the penetration of chitosan into the nuclei of the bacteria and iii) the formation of an external barrier, chelating metals and provoking the suppression of essential nutrients to microbial growth.

Key Words Chitin, cellulose, chitosan.

Introduction

Polysaccharides are the most abundant of the four major classes of biomolecules, which also include proteins, lipids and nucleic acids. They are often classified on the basis of the sequences and linkages between their main monosaccharide components, as well as the anomeric configuration of linkages, the ring size such as furanose or pyranose, configuration of (D- or L-) and any other substituent present. The structural characteristics of chain conformation and intermolecular links control the physicochemical properties of polysaccharides. For example, polysaccharides containing large numbers of hydroxyl groups are often thought of as being hydrophilic. Polysaccharides have been various roles in living organisms, like storage and transport of energy such as glycogen and starch; structural components that is cellulose and chitin.

Chitin

Chitosan, discovered by Rouget in 1859 261, is a technologically important polysaccharide biopolymer. Chemically, it is a high molecular weight linear polycationi cheteropolysaccharide consisting of two monosaccharides, *N*-acetyl-Dglucosamine and D-glucosamine, linked together by *β*- (1→4) glycosidic bonds. Chitosan is a modified natural carbohydrate polymer derived from chitin as a wide range of natural sources like crustaceans, fungi, insects and algae (Tolamite et al., 2000). Commonly, the shell of crustacean consists of protein 30-40%, calcium carbonate and calcium phosphate 30-50% and chitin 20-30% (Knorr, 1984). The second most natural biopolymer of cellulose, chitin (No and Meyers, 1989).

Fiber-like substance of chitosan and a homopolymer of ß-(1→4)-linked N-acetyl-D-glucosamine. Chitin is made up of a linear chain groups while chitosan is obtained by remove sufficient acetyl groups (CH_3-CO) for the molecule to be soluble in acids. Chitin and chitosan is the acetyl content of the polymer.

Chitosan have presence of free amino group is the most useful derivative of chitin (No and Meyers, 1992). Chitosan is a non-toxic, biodegradable polymer of high molecular weight and is very much similar to cellulose, a plant fiber. The only difference chitosan and cellulose is the amine group in the position C-2 of chitosan instead of the hydroxyl group found in cellulose.

Materials and methods

Collection of shrimp shell waste

Shrimp shell waste was collected from the Rameswaram district.

Sample and sampling spot

Shells were removed and thoroughly washed with running tap water with sample care so as to remove impurities, the exoskeleton were subjected to shade drying for 2days for further studies.

Extraction of chitin from shrimp shell

Preparation of chitin

Preparation of chitin was carried in various stages.

- Deproteinization
- Demineralization
- Deacetylation

Deproteinization

For the removal of 40g of shrimp shell was soaked and stride out in 3% solution of NaOH for 5-30 min at 95°c

Demineralization

The preparation of chitin form the shrimp shell waste was followed by diluted HCl solution used for demineralization. Chitin was treated with 4% conc. HCl in order to remove the calcium from shells at 20-24 hrs. The residues washed with deionized water.

Deacetylation

After demineralization the process is carried out with the help of HCl. In deacetylation the shells were washed with water, and treated with 50% NaOH and boiled at 100 C for 2h on a hot plate. The sample is then placed in a room temperature for 30 min. After, the sample were washed with 50% NaOH and filtered in order to retain the solid matter, which is the chitosan. The sample were then uncovered and oven dried at 110°c for 6h. The chitosan obtained will be creamy white color form.

After filtration the residue was washed with distilled water to neutral. Then the residue was immersed with 50% NaOH and boiled at 100ºC for 2hr. And the proteins were removed by filtration method. The deionized water was wash with the residue to neutral. The shrimp shell remains was subjected to the above program for two times. 250 ml of 95% ethanol were sequentially used to remove ethanol-soluble substances from the obtained crude chitin and to dehydrate indicates the formation of creamy white color.

Preparation of Chitosan and Water Soluble Chitosan

10 g was put in to 50 % NaOH at 60ºC for 8h to prepare crude chitosan. After filtration, the residue was washed with hot distilled water at 60ºC for three times. The crude chitosan was obtained by drying in an air oven at 50ºC overnight. Take 1gm of crude chitosan was added 20 ml of 2% (w/w) acetic acid in a water-bath. Conditions were set as 4% H_2O level, 4 hr and 60ºC temperature. After reaction, 10% NaOH was used to pH adjust in neutrality. The residue was removed by filtration method, while twofold volumes of ethanol were added to the filtrate. The crystal of water-soluble chitosan was liberated after

incubation at ambient condition overnight and dried in an air oven at 50°c. The recovery (%) was calculated as (the weight of water-soluble chitosan/the weight of crude chitosan) ×100.

Assay of antibacterial activity of crude a water soluble chitosan

Agar well diffusion

The method is suitable for organism that grows rapidly overnight at 35-37°c. The well is made in medium after inoculation with organisms. When well is loaded with antibiotics, it diffuses in the medium and inhibits the growth of organisms. There is a logarithmic reduction in antibiotic concentration. The zone of inhibition bacterial growth around each well is measured and the susceptibility is determined.

Result and discussion

Production and water soluble chitin form chitinase

The yield of the chitin and the chitosan from the shrimp cell waste was 32%. Chitin was prepared using the alkali treatment the yield of chitin was 40% in the total weight of the dried shells after acetylating the yield of the chitosan was in the range of 56.6%. Where else the chitosan obtained from the shrimp shell was 96.6% (Fig. 1)

Antibacterial activity

Chitosan has a recognized antimicrobial activity, one of the main properties of the polymer. Polysaccharide has antimicrobial action in a great variety of microorganisms, including algae, fungi and bacteria. This glycosaminoglycan has shown antimicrobial activity for some pathogen microorganisms, being highlighted its performance against gram-positive bacteria and various species of yeast. It is assumed that chitosan acts in the cellular wall of the microorganism modifying the electric potential of the cellular membrane. This polysaccharide also acts potentiating other inhibition drugs, as the chlorohexidine gel, once it increases the drug permanence time action place. The antimicrobial activity, the chitosan has demonstrated low toxicity and the resistance development have not occurred. The antimicrobial action of the chitosan and its derivatives suffers influence from factors that depending on the performed role may be classified in four main categories:

1. Microbial factors (microbial species, age of the cell)
2. Intrinsic factors of the chitosan (positive charge density, molecular weight, hydrophobic and hydrophilic characteristics, chelating capacity)
3. Physical state factors (soluble and solid state) and
4. Environmental factors (pH, ionic forces, temperature, time).

In antimicrobial action mechanism the amino groups of the chitosan contact with physiological fluids are prorogated and bind to anionic groups of the microorganisms, resulting in the agglutination of the microbial cells and inhibition of growth. On the other hand interacting with the bacterial cell, the chitosan, promotes displacement of Ca++ of the anionic sites of the membrane resulting in cell damages

The chitosan exhibited the maximum zone of inhibition against E.coli, Klebsiella, Staphylococcus aureus, Streptococcus shown in table.

Chitosan is a versatile material with proved antimicrobial activity. Three antibacterial mechanisms have been proposed: i) the ionic surface interaction resulting in wall cell leakage; ii) the inhibition of the mRNA and protein synthesis via the penetration of

chitosan into the nuclei of the microorganisms; and iii) the formation of an external barrier, chelating metals and provoking the suppression of essential nutrients to microbial growth. It is likely that all events occur simultaneously but at different intensities. In general the lower the MW and the DA, the higher will be the effectiveness on reducing microorganism growth and multiplication. Chitosan has higher activity on gram-positive or on gram-negative bacteria. Water soluble derivatives, which can be attained by chemical introduction of CH_3 in the main chain, enhancing the chitosan applicability in a large pH range and also improve the antimicrobial activity.

Table 1: Antibacterial activity of the crude chitosan against *E. coli*

Concentration	Well diameter	Zone of Inhibition	Inhibition length
0.5 %	0.8	2.0	1.2
1.0 %	0.8	2.3	1.2
1.5 %	0.8	2.1	1.2
2.0 %	0.8	2.0	1.2

Table 2: Antibacterial activity of the water soluble chitosan against *E. coli*

Concentration	Well diameter	Zone of Inhibition	Inhibition length
0.5 %	0.8	2.1	1.3
1.0 %	0.8	2.3	1.5
1.5 %	0.8	2.4	1.6
2.0 %	0.8	2.0	1.2

Table 3: Antibacterial activity of the crude chitosan against *Klebsiella*

Concentration	Well diameter	Zone of Inhibition	Inhibition length
0.5 %	0.8	2.0	2.2
1.0 %	0.8	2.3	2.8
1.5 %	0.8	2.8	2.0
2.0 %	0.8	2.1	2.2

Table 4: Antibacterial activity of the water soluble chitosan against *Klebsiella*

Concentration	Well diameter	Zone of Inhibition	Inhibition length
0.5 %	0.8	2..4	1.9
1.0 %	0.8	2.0	1.7
1.5 %	0.8	2.1	1.5
2.0 %	0.8	2.2	1.4

Table 5: Antibacterial activity of the crude chitosan against *Staphylococcus aureus*

Concentration	Well diameter	Zone of Inhibition	Inhibition length
0.5 %	0.8	2.6	1.5
1.0 %	0.8	2.3	1.0
1.5 %	0.8	2.2	1.7
2.0 %	0.8	2.1	1.9

Table 6: Antibacterial activity of the water soluble chitosan against *Staphylococcus aureus*

Concentration	Well diameter	Zone of Inhibition	Inhibition length
0.5 %	0.8	2.3	1.8
1.0 %	0.8	2.9	1.7
1.5 %	0.8	2.5	1.2
2.0 %	0.8	2.2	1.5

Table 7: Antibacterial activity of the crude chitosan against *Streptococcus pneumonia*

Concentration	Well diameter	Zone of Inhibition	Inhibition length
0.5 %	0.8	2.5	1.5
1.0 %	0.8	2.4	1.4
1.5 %	0.8	2.1	1.7
2.0 %	0.8	2.6	1.0

Table 8: Antibacterial activity of the water soluble chitosan against *Streptococcus pneumonia*

Concentration	Well diameter	Zone of Inhibition	Inhibition length
0.5 %	0.8	2.4	1.7
1.0 %	0.8	2.1	1.6
1.5 %	0.8	2.6	1.4
2.0 %	0.8	2.9	1.2

Fig. 1: Antibacterial activity of the crude chitosan and water soluble chitosan from shrimp shell wastes

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